Agustin Freire

+54(2266)485600 ◊ Argentina, Buenos Aires, Balcarce
agustinfreire2002@outlook.com.ar ◊ https://www.linkedin.com/in/agustin-freire-4770b3352/ ◊ https://github.com/ricotero2002

OBJECTIVE

Software Engineer with experience in AI development and intelligent systems, specializing in LLM integration, agent development, and AI deployment and RAG pipelines. Successfully led the end-to-end development of an educational platform that is currently operational. I am passionate about creating functional products that solve problems and meet customer needs. I seek a challenging and learning environment to apply and expand my knowledge.

EDUCATION

Software Engineer Degree at National University of Central Buenos Aires (**UNICEN**). 2021 - 2026.
GPA: 8.86

SKILLS

Languages	Python, Java, JavaScript,, C#, C++
AI, MLOPS	Generative AI, NLP, LLMs (Gemini, OpenAI), LangChain, LangGraph, NeMo Guardrails, Advanced RAG, Prompt Engineering, Multi-Agent Systems, HybridSearch, Knowledge Graphs, LangSmith, PostgreSQL, Neo4j, DeepEval, Celery, Apache Kafka, Garak, ChromaDB, Pydantic, Redis, SQLite, MCP, API, Cloud Native, PyTest.
DevOps	Docker, Uvicorn, Github Actions, Webhooks, MinIO, RabbitMQ, Kubernetes .
Other	Spring Boot, FastAPI, Django, Nextjs, React Native, Agile (Scrum, Jira, ClickUp), SAFe, Unity3D.

EXPERIENCE

Software Engineer - OOPingo (Final Project) **|** ISISTAN | October 2025 – December 2025 | Argentina (Remoto)

- **Led** the end-to-end development of a gamified educational platform, managing the full SDLC from architectural design to deployment.
- **Designed** an Client-Server architecture using **React Native Expo**, **Java (Spring Boot)** and **PostgreSQL**, ensuring scalability for user and content management.
- **Implemented** asynchronous messaging pipelines with **RabbitMQ**, optimizing system performance and decoupling critical services.
- **Deployed** the infrastructure using **Docker** containers and **Nginx**, ensuring consistent environments across development and production.

Teaching Assistant (Ayudante de Cátedra) UNICEN - Facultad de Ciencias Exactas | March 2024 – February 2026 | Tandil, Arg

- Provided technical mentoring in Object-Oriented Programming and Information Theory, guiding students in software design, data structures, and complex algorithms.

PROJECTS

AI ENGINEERING & R&D PROJECTS

AI Client Framework (Unified LLM Orchestrator).

- **Engineered** a production-grade Python framework orchestrating multiple providers (OpenAI, Gemini), ensuring **100% uptime** via automatic fallbacks.
- **Integrated Pydantic** for structured output validation and intelligent prompt caching, significantly reducing API costs.

Human-in-the-Loop IDL Agent (MCP Implementation).

- **Built** a secure agentic workflow using **LangGraph** and the **Model Context Protocol (MCP)** for safe terminal automation.
- **Designed** robust security guardrails, human approval gates, and **automatic rollback** capabilities for critical operations.

AI-EventStream - Distributed AI Architecture.

- **Designed** a scalable backend for asynchronous AI processing using **FastAPI**, **Apache Kafka**, and **Celery.**
- **Implemented** a Semantic Cache using **Redis Stack** (Vector Search), optimizing response latency to **<50ms**.

Gbeder Multi-Agent Research System.

- **Designed** multi-agent pipelines comparing orchestration patterns (Sequential vs. Supervisor vs. Reflexion) using **LangGraph**.
- **Integrated** custom search tools and **DeepEval** frameworks to measure and improve report grounding and accuracy.

NeMo Defense Bot.

- **Implemented** a multi-layered security system with **NVIDIA NeMo Guardrails** to block jailbreaks and prompt injections.
- **Achieved** minimal risk scores in adversarial testing using **Garak** probes, effectively preventing PII leakage.

Rag Multi-Strategy Project.

- **Conducted** comparative research on four RAG strategies (Naive, Advanced, Agentic, Graph RAG) using a custom dataset.
- **Analyzed** trade-offs using **DeepEval**, demonstrating that **Graph RAG** offered the highest faithfulness and response quality.

KEY ACADEMIC PROJECT

Person Tracking System

- Developed a real-time person tracking application in a team of 7, utilizing Scrum methodology for agile delivery.
- Built a cross-platform desktop application that integrates YOLO object detection with a high-performance backend.
- Implemented a reactive frontend for data visualization, utilizing WebSockets to ensure low-latency communication.